

RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

Department of Cyber Security

2026-2027- Odd Semester

III Year /V Sem

CS23533-FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

Zeroth Review – **31.07.2026**

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PROJECT TITLE

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OUTLINE

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ABSTRACT

Large Language Models (LLMs) such as OpenAI GPT, Google Gemini, Anthropic Claude, and Meta Llama have become integral to enterprise applications including customer support, software development, finance, healthcare, and education. However, organizations face challenges such as repeated AI requests, increasing API costs, inconsistent responses, limited governance, and lack of centralized monitoring.

SentinelAI is an Enterprise AI Gateway that provides centralized management, governance, and intelligent optimization for AI-powered applications. Instead of allowing applications to communicate directly with LLMs, SentinelAI acts as an intermediary that authenticates users, manages organizations, enforces policies, monitors AI usage, and maintains audit logs.

The proposed system introduces an Enterprise Memory Intelligence module that stores verified AI responses, reducing API calls, response time, token usage, and operational cost. Developed using Django, Django REST Framework, React, and PostgreSQL.

OBJECTIVE

- Develop a centralized AI Gateway for enterprise applications.
- Reduce unnecessary LLM API calls using Enterprise Memory Intelligence.
- Improve response consistency by reusing verified AI responses.
- Reduce API costs and token consumption.
- Provide organization-based AI management.
- Monitor AI usage through centralized analytics.
- Maintain audit logs for all AI interactions.
- Support secure and scalable AI integration.

Literature Survey

TITLE	YEAR	AUTHOR	TECHNIQUE
Retrieval-Augmented Generation for Knowledge-Intensive NLP	2020	Patrick Lewis ,Ethan Perez, Aleksandra Piktus et al.	Retrieval-Augmented Generation (RAG)
Prompt Injection Attacks Against Large Language Models	2023	Simon Willison	Prompt Injection Analysis
LangChain: Building LLM Applications	2023	Harrison Chase	LLM Orchestration Framework

EXISTING SYSTEM

- Applications directly talk to Language Model providers.
- Every time you give a prompt it makes a request, to the API.
- This can get expensive because high API usage means operational costs.
- When you ask the question multiple times you get the same response multiple times.
- There is not a way to keep track of how Language Models are being used in a centralized manner.
- Companies do not have a way to control and analyze how Language Models are being used.
- Language Models do not save responses to use again across applications.

PROPOSED SYSTEM

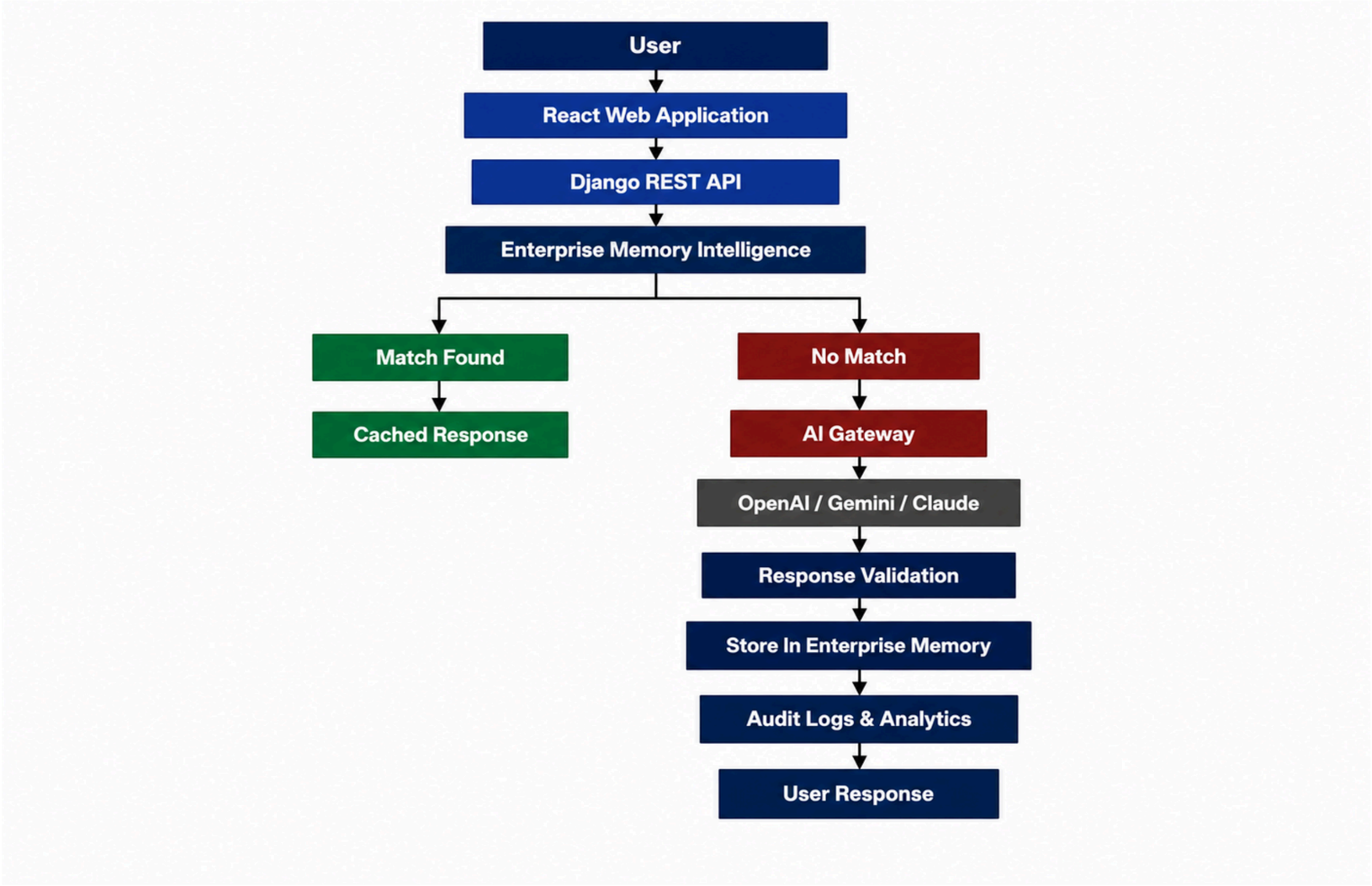
SentinelAI acts as a centralized AI Gateway between enterprise applications and LLM providers. Every user request first passes through the Enterprise Memory Intelligence module. If a similar verified response already exists, the system returns the cached response without contacting the LLM. Otherwise, the request is forwarded to the selected LLM, and the generated response is validated, stored, and reused for future requests.

The platform additionally provides authentication, organization management, audit logging, AI usage analytics, cost monitoring, and centralized governance through an interactive dashboard.

Novelty in Proposed System

- Enterprise Memory Intelligence for response reuse.
- Reduction of redundant LLM API calls.
- Lower operational costs through intelligent caching.
- Consistent responses using verified AI outputs.
- Centralized AI Gateway for multiple enterprise applications.
- AI usage analytics and cost monitoring dashboard.
- Organization-based AI governance and management.
- Scalable architecture supporting multiple LLM providers.

BLOCK DIAGRAM(PROPOSED SYSTEM)



Hardware/Software Requirements

Hardware Requirement:

- Intel Core i5 Processor or above
- 8 GB RAM (Minimum)
- 256 GB SSD Storage
- Stable Internet Connection

Software Requirement:

- Windows 11
- Python 3.13+
- Django & Django REST Framework
- React.js / Bun
- SQLite
- Postman
- Git & GitHub
- Gemini API

References

- **[1]** Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020.
- **[2]** Willison, S. (2023). Prompt Injection Attacks Against Large Language Models. Personal Blog / ACL Workshop.
- **[3]** Chase, H. (2023). LangChain: Building LLM Applications. GitHub / LangChain Documentation.
- **[4]** OpenAI. (2023). GPT-4 Technical Report. arXiv:2303.08774.
- **[5]** Anthropic. (2022). Constitutional AI: Harmlessness from AI Feedback. arXiv:2212.08073.